Solution to exercises in Lecture 2

Q1) Conceptually the change is minimal: Rather than using Eq. (4) in the lecture note, we should use Eq. (5) in the line that says

" Compute Eq. ... for all i, j"

In practice, there is a subtlely in implementation for python. For Jacobi iteration you can use array manipulation which is much faster. For Gauss-Sidel you can only use double for loop over i, j, which is slower.

Q2) We need to balance accuracy vs cost.

Any answer between $\frac{1}{40}$ and 0.00625 is reasonable.

The error is small and the cost has not increased steeply.